THE LOCUS WHERE A MAP IS UNRAMIFIED IS OPEN INSIDE THE LOCUS WHERE IT IS UNIVERSALLY INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 11586-11639, immediately following the statement at 11565-11584.

Editorial note. Five cross-chapter pointers (Morphisms of Spaces, Algebraic Spaces) plus a same-chapter companion are recorded in assembly.external_references.

Editorial note. The proof produces an open subspace and verifies the factorisation property; both steps are inside the counted span.

Allowed standard prerequisites (editorial).

- The unramified locus of a morphism. chapter lemma lemma-where-unramified, cited in the proof. Supplies the corresponding statement for unramifiedness which is adapted here to the universally injective setting.
- Criterion for universal injectivity. Morphisms of Spaces, lemma spaces-morphisms-lemma-universally-injective, cited in the proof. Turns universal injectivity into a condition on the diagonal, which is what the closedness hypotheses are used to test.
- The diagonal of an unramified morphism. Morphisms of Spaces, lemma spaces-morphisms-lemma-diagonal-unramified-morphism, cited in the proof. Relates the unramified locus to openness of the diagonal.
- Open subspaces and factorisation through them. Algebraic Spaces, lemmas spaces-properties-lemma-open-subspaces, spaces-properties-lemma-factor-through-open-subspace and spaces-properties-lemma-points-cartesian, cited in the proof. Convert the set-theoretic conditions obtained in the proof into an open subspace with the required factorisation property.

Source reference. Original lemma 05X8: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 11529–11529. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 05W1: focused-seeds/geometry/S9-spaces-unramified-univ-inj/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 040X: focused-seeds/geometry/S9-spaces-unramified-univ-inj/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

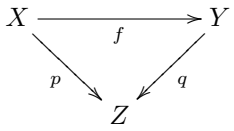
Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 03IE: focused-seeds/geometry/S9-spaces-unramified-univ-inj/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03BZ: focused-seeds/geometry/S9-spaces-unramified-univ-inj/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-spaces-unramified-univ-inj/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Consider a commutative diagram*



of algebraic spaces. Assume that

(1) *p is locally of finite type,*
(2) *p is closed, and*
(3) *$p_2 : X \times_Y X \to Z$ is closed.*

Then there exists an open subspace $W \subset Z$ such that a morphism $Z' \to Z$ factors through W if and only if the base change $f_{Z'} : X_{Z'} \to Y_{Z'}$ is unramified and universally injective.

Proof. After replacing Z by the open subspace found in Lemma 05X8 we may assume that f is already unramified; note that this does not destroy assumption (2) or (3). By Morphisms of Spaces, Lemma 05W1 we see that $\Delta_{X/Y} : X \to X \times_Y X$ is an open immersion. This remains true after any base change. Hence by Morphisms of Spaces, Lemma 040X we see that $f_{Z'}$ is universally injective if and only if the base change of the diagonal $X_{Z'} \to (X \times_Y X)_{Z'}$ is an isomorphism. Let $W \subset Z$ be the open subspace (see Properties of Spaces, Lemma 03BZ) with underlying set of points

$$|W| = |Z| \setminus |p_2| \left(|X \times_Y X| \setminus \mathrm{Im}(|\Delta_{X/Y}|) \right)$$

i.e., $z \in |Z|$ is a point of W if and only if the fibre of $|X \times_Y X| \to |Z|$ over z is in the image of $|X| \to |X \times_Y X|$. Then it is clear from the discussion above that the restriction $p^{-1}(W) \to q^{-1}(W)$ of f is unramified and universally injective.

Conversely, suppose that $f_{Z'}$ is unramified and universally injective. In order to show that $Z' \to Z$ factors through W it suffices to show that $|Z'| \to |Z|$ has image contained in $|W|$, see Properties of Spaces, Lemma 03IE. Hence it suffices to prove the result when Z' is the spectrum of a field. Denote $z \in |Z|$ the image of $|Z'| \to |Z|$. The discussion above shows that

$$|X_{Z'}| \longrightarrow |(X \times_Y X)_{Z'}|$$

is surjective. By Properties of Spaces, Lemma 03H4 in the commutative diagram

$$\begin{array}{ccc} |X_{Z'}| & \longrightarrow & |(X \times_Y X)_{Z'}| \\ \downarrow & & \downarrow \\ |p|^{-1}(\{z\}) & \longrightarrow & |p_2|^{-1}(\{z\}) \end{array}$$

the vertical arrows are surjective. It follows that $z \in |W|$ as desired. □